

Basic Code Analysis: JVM Memory OutOfMemoryError Diagnostics

Duration : 90 Min.

Scenario

Your server crashed with a java.lang.OutOfMemoryError: Java heap space. Analyze the JVM heap memory dump logs and thread logs to find the memory leak.

Target File Location & Creation

File to Create/Update: student_workspace/solution.json **Input Resource File to Inspect:** student_workspace/jvm_heap_dump.txt

Task Objectives

Perform the following actions inside the student_workspace directory: - Document the leak suspect object class name and line number inside 'solution.json'.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect jvm_heap_dump.txt inside student_workspace/.
3. Open solution.json in student_workspace/ and perform the required modifications.
4. Save your changes (Ctrl + S or Cmd + S).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your progress by running python run.py locally in the terminal.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Test Case	Requirement	Marks
TC1	solution.json exists	3 Marks
TC2	Memory leak class suspect correctly identified	3 Marks

TC3	Correct list/collection object causing leak identified	3 Marks
TC4	Line number of insertion leak mapped correctly	3 Marks
TC5	RCA analysis notes populated	2 Marks
TC6	Recommended heap memory increase parameter included	2 Marks
TC7	Thread dump correlation correct	2 Marks
TC8	JVM arguments settings validated	2 Marks

Total Score: 20 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.